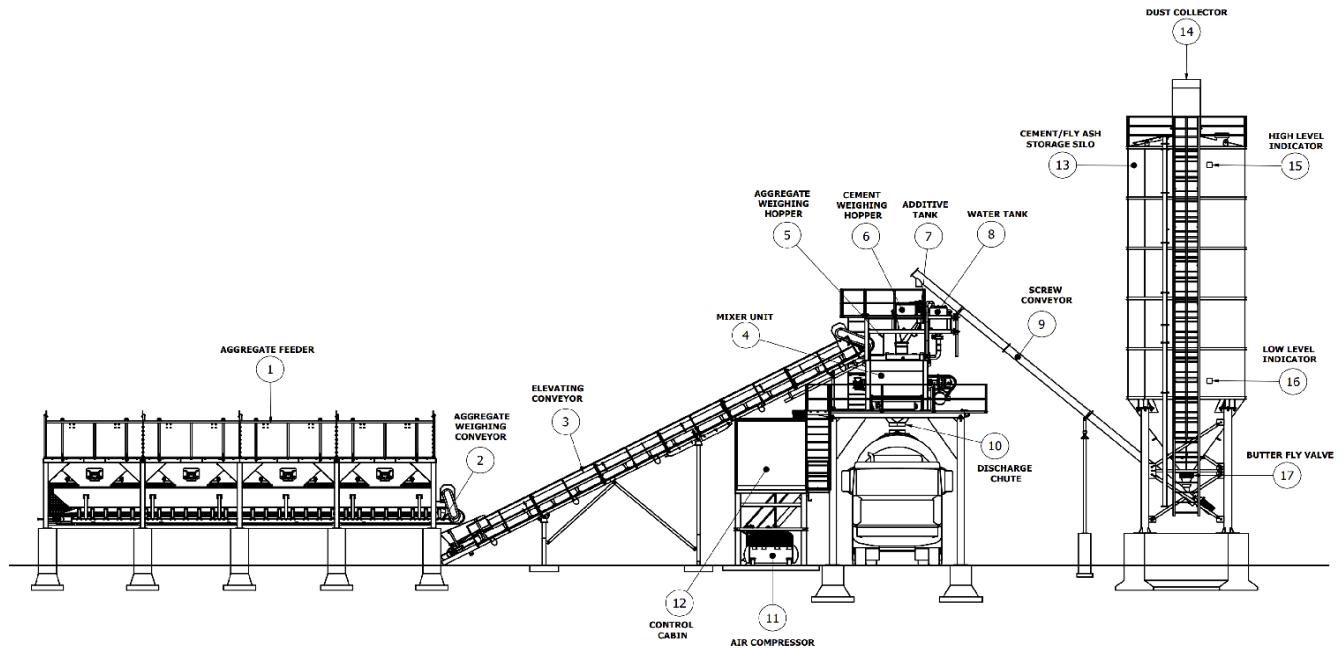




STATIONARY CONCRETE BATCHING PLANT CAP. 30 TO 200 M³



DETAIL FORM FOR STATIONERY CONCRETE BATCHING PLANT WITH TWIN SHAFT MIXER

BELOW REFERENCE IS 180 M³ PER HOUR

- | | |
|--------|---|
| Item 1 | Four/Five (4/5) bin feeder with individual bin , conveyors, weighing/mixing section & control cabin |
| Item 2 | 150 tons cement silo with accessories (Optional) |
| Item 3 | cement feeding system for cement silo(Optional) |
| Item 4 | Plant control system |
| Item 5 | Wiring |
| Item 6 | Supervisory installation and commissioning |

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TECHNICAL SPECIFICATION

PLANT CAPACITY

For Reference: 180 m³ based on a mixer capacity of 7500 / 5000 litres. Of vibrated concrete per batch 5.0 m³ and at (95) second weigh/mix cycle.

Assuming the following conditions: -

- 1) 100% Plant utilisation
- 2) Material density (2400) Kg/m³

1 INDIVIDUAL 4/5 BIN FEEDERS, GATHERING CONVEYORS, DISCHARGE CONVEYORS & CABIN

1.1 AGGREGATE HOPPERS

Feeders	-	Four/Five (4/5) individual Bin
Capacity	-	25 m ³ /each bin
Loading width	-	2.9 meter
Loading height	-	6.5 meter
Material	-	5 mm thick mild steel plate
Vibrator	-	Two
Discharge gates	-	Eight (Each bean two) pneumatically controlled discharge gates.

1.2 GATHERING / WEIGH CONVEYOR

Weigh conveyor	-	Mounted under weigh hoppers
Length	-	13 Metres
Belt	-	1000 mm wide 4 ply with vulcanised joint
Idlers	-	Troughed bolted to steel section support frame
Drive	-	30 HP gear motor unit direct on head shaft
Belt tensioning	-	Tension bolts fitted to tail drum slide bearings
Belt scraper	-	Fitted under head drum
Weigh hopper	-	Manufactured from 5 mm mild steel plate
Weighing capacity	-	6.5 m ³ total from all hoppers, mounted on load cells
Load cells	-	Four (6)
Transfer load capacity	-	250 Kg. cm ²
Emergency stop switch-	-	One fitted at the bottom of the weigh conveyor



1.3 CHARGING CONVEYOR

Charging conveyor	-	Mounted between four bin feeders and mixing tower.
Length	-	15 Metres
Belt	-	1000 mm wide 4 ply chevron type (Ribbed) with vulcanised joint
Idlers	-	Troughed bolted to steel section support frame
Drive	-	40 HP gear motor unit direct on head shaft
Belt tensioning	-	Tension bolts fitted to tail drum slide bearings
Belt scraper	-	Fitted under head drum
Transfer load capacity		250 Kg. cm ²

1.4 CEMENT WEIGH HOPPERS

Capacity	-	One 2700 Kg. load cell mounted
Load cells	-	1800 kg. Shear beam type - 3 no.
Plate thickness	-	5 mm steel plate
Discharge doors	-	300 mm pneumatically operated butterfly valves with On-Off limit switch
Vibrator	-	0.33 HP vibrating motor
Total inlets	-	Two for optional customer supply flyesh silo / feed screw

1.5 WATER WEIGH HOPPER

Capacity	-	1500 Kg. load cell mounted
Load cells	-	1000 kg. "S" type - 3 no.
Plate thickness	-	4 mm mild steel plate
Discharge valve	-	100 mm pneumatically operated butterfly valve with On-Off switch

1.6 ADDITIVE WEIGH HOPPER

Capacity	-	20 Kg. load cell mounted
Load cells	-	50 kg. "S" type - 1 no.
Material	-	5 mm thick and transparent acrylic dispenser.
Discharge valve	-	Spherical valve operated via actuator

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1.7 TWIN SHAFT MIXER (7500/5000) – SICOMA MAKE OR BHS

- VIBRATED MATERIAL YIELD 5000 LITRES PER CYCLE
- WITH AUTOMATIC DISCHARGE DOOR, HYDRAULIC OPENING
- WITH EMERGENCY MANUAL PUMP FOR DISCHARGE DOOR OPENING
- WITH MIXING BLADES IN WEAR RESISTING CAST IRON (600HB)
- WITH SCRAPER BLADES IN WEAR RESISTING CAST IRON (600HB)
- WITH BOTTOM LINERS IN WEAR RESISTING CAST IRON (16mm TH) (500HB)
- WITH LATERAL LINERS IN WEAR RESISTING CAST IRON (12mm TH) (500HB)
- HYDRAULIC POWER PACK MOTOR 1 x 2.25
- AUTOMATIC GREASE PUMP
- DISCHARGE GATE THREE STAGES LIMIT SWITCHES
- INSPECTION HATCH WITH MESH GRILL AND SAFETY SWITCH
- JUNCTION BOXES FOR POWER SUPPLY AND CONTROL
- MIXING BLADE, 20 SETS
- SCRAPER BLADE, 4 SETS
- MIXING ARM, 20 SETS
- ARMS 90 DEGREE
- SANDBLASTING, UNDERCOAT PRIMER AND FINISH PAINT
- SINGLE OR DOUBLE ELECTRIC MOTOR 415 V, 3PH, 50HZ
- PLATFORM STEP

1.8 PNEUMATICS

- | | | |
|------------|---|--|
| Compressor | - | 10 HP to give 12.3Kg. Pressure with 45.8 CFM of piston displacement. |
| Pneumatics | - | Cylinder, Solenoid valves, nylon pipe work and fittings |

1.9 CONTROL CABIN

- | | | |
|--------------|---|---|
| Type | - | Fixed control cabin with four legs. |
| Construction | - | Corrugated sheet steel cladding and folds into under frame for transport |
| Base frame | - | Constructed from rolled steel section |
| Roof | - | Sheet steel. Roof void fully insulated with 50mm thermocol sheet. |
| Floor | - | Wooden floor with carpet. |
| Walls | - | Clad internally with decorative boarding. Walls fully insulated with 50mm mineral wool. |
| Windows | - | Three sides |
| Lighting | - | Overhead LED fitted. |
| Heating | - | Air conditioning unit (Optional) |
| Door | - | Wide access door with door closer and lock |
| Power | - | Access from bottom of panel |

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1.10

PLATFORMS

Platforms - Maintenance platform for ease of access to mixer and weigh hoppers, with hand railing and access ladders

1.11

UNDER FRAME

Under frame - Fabricated rolled steel section chassis
Support jacks - Fixed supports under mixer
Loading height - 4.2 meter clearance under mixer outlet chute

2

150 T CEMENT SILO (OPTIONAL)

2.1 100 Ton Capacity Silo for Cement With fabricated 4 Column, Cross Bracings, Railing at Top, Climbing Ladder, Seamless Pipe (Dia: 100 mm Seamless pipe for feeding material in silo), Aeration System at Bottom of cone & with other necessary filaments.

2.2

SILO ACCESSORIES (1 set)

Double Flange Butterfly Valve with Cast Iron Disc Dia. 300 mm and Manual Lever for above Butterfly valve	- One unit
Special Dust Collector for Silo Venting	- One unit
Level Indicator 115/230 V/AC Paddle type Rotating	- Two units
Pressure Safety Relief Valve M.O.C. – M– M.S.	- One unit
Vibrating Bin aerator - (1 set inclusive of below items)	
1" Two way valve with coil	- 1 No.
1" Filter regulator with gauge	- 1 No.
Bracket for FRL	- 1 No.
Manifold 1/2" female x 2 outlet.	- 1 No.
Reducer 1" male X 1/2" female.	- 1 No.
1" Nipple-- 3 Nos	
Straight connector 1/2" x 12mm OD	- 4 No.
Equal tee 12mm OD	- 2 No.
PU tube 12mm OD	- 10 Meter
Male elbow 3/8" X 12mm OD	- 4 No.

Transfer screws - 323 mm diameter with 10 meters length cement screw conveyor

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3 **SILO FEEDING SYSTEM (OPTIONAL)**

3.1 Cement feeding compressor for bulker feeding

SPECIFICATION	
CAPACITY (FAD)	400(800 M3 / Hr)
PRESSURE	1.0 Kg. / Cm2.
MACHINE SPEED	1450 RPM
POWER AT SHAFT	37.5 K.W.
4 RECOMM. MOTOR	50 H.P. / 1450 RPM.
TYPE OF DRIVE	V – BELT
TYPE OF COOLING	AIR COOLED

CONTROLS

- 4.2 **CONTROL DESK** - Containing key switches for manual/auto control, manual Start/stop buttons and PLC weigh/mix control system
- 4.3 **WEIGH/MIX** - Weighing and mixing control system comprising of:
Siemens PLC operator interface terminal with Sunlight Visible Colour FT (7 INCH) Touch Screen Super Bright
Display ensures maximum readability with a clear keyboard design and eight multifunction soft keys. Automatic control of the batching process via English interface with unlimited recipe storage. Printer supplied for production logging of essential data from each batch. Production data stored on non-volatile Compact Flash, accessible by USB cable. Network port to allow remote viewing and control of the batching process on customers PC.

OPTIONAL

SCADA type control system with Wi-Fi internet attachment facility and also facility to operate plant by mobile / tablet.

5 **WIRING**

Feeder and mixing unit section of stationary concrete plant is prewired with junction box. Cable from junction box to control panel and from panel to genet is on customer's scope

GUARDS

Safety guards are provided over all V belt drives, chain drives and gears.

STEELWORK

All welds to be cleaned as necessary, steelwork to be wire brushed and generally cleaned of all mill scale etc before painting.

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PAINTING

All external surfaces are chipped and painted with one-coat primer with double coated PU paint.

VOLTAGE

400 Volt, 3 phase, 50 Hz

Stationery Concrete Plant Motor List

Item	Quantity	HP	Qty.	Power (HP)
1	Twin shaft mixer	125	2	250
2	Weigh conveyor	30	1	30
3	Agg hopper vibrator	0.5	1	0.5
4	Charging Conveyor	40	1	40
5	Cement weigh hopper vibrator	0.33	1	0.33
6	Air compressor	10	1	10
7	Water pump	7.5	1	7.5
8	Additive motor	2	1	2
9	Hydraulic power pack	3	1	3
10	Screw conveyor 323 mm x 12 meters	30	2	60
11	Cement feeding system (Optional)	50	1	50
			TOTAL	410 HP

Sr. No.	DESCRIPTION	MAKE
1	Mixer gear box	Brevini, Bonfiglioli, Bonfiglioli, PBL, Top Gear
2	All other gear boxes	Bonfiglioli/Wam/Prashant/Transolo/Top Gear
3	Screw conveyor	WAM / IVA BULK
4	Motors	Siemens / Hindustan / Crompton
5	Conveyor belt	MRF OR COMPERABLE
6	V belt	Fenner
7	Compressor	Anest Iwata OR Indo Air
8	Pneumatics	SMC
9	Load cell	SENSORTRONICS OR ASI OR KELLY
10	Pedestal bearing	TR
11	Vibrating motors	ISI Standard
12	Silo accessories	IVA BULK/ WAM
13	Water Pump	Kirloskar or kebee



Note: We recommend you to buy gen set of 380 kva. For the plant

Main cable size: 300 sq. mm x 3 ½ core armoured copper – Length depending on the Generator distance.

Exclusions

Civil works, foundations etc

Mains power supply or generator (for usage of 380-400 KVA , 3 phases for grid and equivalent generator for power consumption on site)

Mains incoming cable or distribution board

Water supply and incoming pipework

Aggregate storage hopper grids

Site labour and carnage

Any item not specified

Warranty

The equipment is covered by guarantee under proper use, for faulty material or workman ship for a period of 12 month from the date of dispatch.

Consumable, electric components, rubber components and general wear and tear components are not covered under guarantee.

Purchaser shall inform supplier in writing, of any defect in equipment noticed during guarantee period. On receipt of purchaser's letter, supplier shall at his option, replace or repair free of cost the defective part supplied by him. In case repair is required to be done at works, purchaser shall send such defective part to the works after receiving supplier's confirmation.

We are in no event be liable for any damages whatsoever (including special, incidental consequential, direct or indirect damages, for personal injury, loss of business profits, or any other pecuniary loss) arising out of the use of or inability to use the equipment.

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